WORKED SOLUTION

STEP I 2011, Question 9

Mechanics • Projectile motion

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Read the question before you go on. It is in the STEP Support Programme's question database:

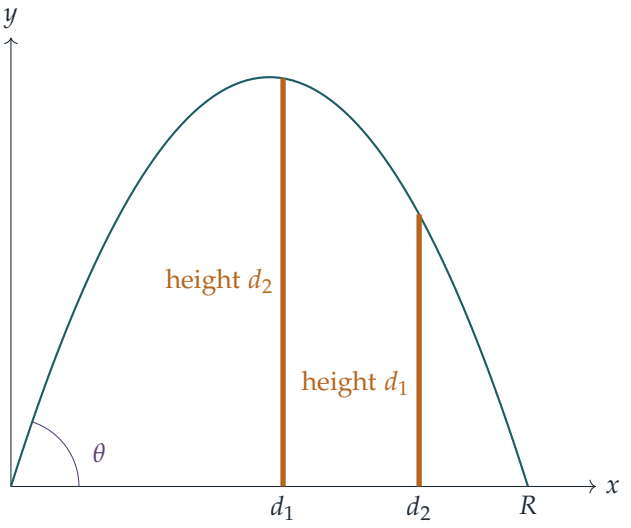
step.maths.org/questions/11-s1-q9

The key idea

Write the trajectory as a single equation $y = x\tan\theta - ax^2$, where the constant a packs together g, the speed and the angle. "Just passes over" a wall means the path goes through the top corner of the wall, so each wall gives one linear equation in the two unknowns $\tan\theta$ and a. Solving that pair gives the angle, and the range is then just the other root of $y = 0$.

Solution

We model the particle as a point moving under gravity alone (no air resistance), projected from the origin O with speed u at angle θ, with x horizontal and y vertically up. The walls have different heights, so we assume $d_1 \neq d_2$.



(The picture uses $d_1 < d_2$; nothing in the algebra depends on which is larger.)

The trajectory

At time t we have $x = ut\cos\theta$ and $y = ut\sin\theta - \frac{1}{2}gt^2$. Eliminating $t = x/(u\cos\theta)$,

$$y = x\tan\theta - ax^2, \qquad \text{where } a = \frac{g}{2u^2\cos^2\theta} > 0.$$

Write $T = \tan\theta$. The particle just clears each wall, so it passes through the tops (d_1, d_2) and (d_2, d_1):

$$d_2 = d_1 T - a d_1^2, \tag{1}$$
$$d_1 = d_2 T - a d_2^2. \tag{2}$$

Finding $\tan\theta$

To eliminate a, multiply (1) by d_2^2 and (2) by d_1^2, so that the a terms are both $a d_1^2 d_2^2$:

$$d_2^3 = d_1 d_2^2\, T - a d_1^2 d_2^2,$$
$$d_1^3 = d_1^2 d_2\, T - a d_1^2 d_2^2.$$

Subtracting,

$$d_2^3 - d_1^3 = d_1 d_2 (d_2 - d_1)\, T.$$

Ask yourself Can we divide by $d_2 - d_1$?

Yes: the walls have different heights d_2 and d_1, so $d_1 \neq d_2$. (If they were equal there would be only one wall, and one point cannot fix both θ and u.) Since $d_2^3 - d_1^3 = (d_2 - d_1)(d_2^2 + d_1 d_2 + d_1^2)$, dividing by $d_1 d_2 (d_2 - d_1)$ gives

$$\tan\theta = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 d_2},$$

as required.

The range

The particle lands where $y = 0$ with $x \neq 0$, that is at $x = T/a$. So we need a as well. Dividing (1) by d_1 and (2) by d_2,

$$\frac{d_2}{d_1} = T - a d_1, \qquad \frac{d_1}{d_2} = T - a d_2.$$

Subtracting the first from the second removes T:

$$a(d_2 - d_1) = \frac{d_2}{d_1} - \frac{d_1}{d_2} = \frac{(d_2 - d_1)(d_2 + d_1)}{d_1 d_2}, \qquad \text{so} \qquad a = \frac{d_1 + d_2}{d_1 d_2},$$

again using $d_1 \neq d_2$. Therefore the range is

$$R = \frac{T}{a} = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 d_2} \cdot \frac{d_1 d_2}{d_1 + d_2} = \frac{d_1^2 + d_1 d_2 + d_2^2}{d_1 + d_2}.$$

As a sanity check, $R - d_2 = \dfrac{d_1^2}{d_1 + d_2} > 0$ and similarly $R - d_1 = \dfrac{d_2^2}{d_1 + d_2} > 0$, so the particle lands beyond both walls, as it must.

? Ask yourself Where did g and u go?

They only ever appear through the single combination a, and the two walls fix a completely: $u^2 = g/(2a\cos^2\theta)$. So the walls determine the whole trajectory, and the range depends on d_1 and d_2 alone.

Checks

- Symmetry: swapping d_1 and d_2 just swaps the walls, and both answers are symmetric in d_1, d_2.
- Numbers: with $d_1 = 2$, $d_2 = 3$ we get $\tan\theta = \frac{19}{6}$, $a = \frac{5}{6}$ and $R = \frac{19}{5}$. Then $y(2) = \frac{19}{3} - \frac{10}{3} = 3$ and $y(3) = \frac{19}{2} - \frac{15}{2} = 2$, as required. Stepping the motion numerically with $g = 9.8$ also lands at $x = 3.8$.
- Since $d_1^2 + d_2^2 \geq 2d_1 d_2$, we get $\tan\theta \geq 3$, so the launch is always steep (at least about $71.6°$). That makes sense: to clear the taller wall first and then come down to the shorter one so soon, the path has to be a narrow arch.

Common slips

- Mixing up which wall has which height: the wall at distance d_1 has height d_2.
- Writing a with $\cos\theta$ instead of $\cos^2\theta$, or not treating a as a single unknown, which leaves a messy system in u and θ.
- Dividing by $d_2 - d_1$ without saying why it is non-zero.

Beyond STEP

The trick here is that a family of curves with two free parameters (here $\tan\theta$ and a) is pinned down by two points it must pass through, a small case of interpolation. The same idea of choosing the right combination of unknowns appears throughout mechanics, for example in finding the "envelope" of all trajectories with a fixed launch speed (the parabola of safety). For more projectile and particle mechanics at first-year level, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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